

3.83-Tb/s Single-Wavelength Coherent Transmission Enabled by 450-GBaud All-Electronic I/Q Signaling

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Abstract We report a single-wavelength coherent transmission with 3.83-Tb/s net bit rate over 10-km SSMF. This is enabled by 450-GBaud all-electronic signaling using a pair of 226-GHz digital-band-interleaved DACs driving a thin-film LiNbO₃ I/Q modulator. ©2026 The Author(s)

Introduction

The need to reduce cost, power consumption and management complexity per bit drives sustained scaling of the per-wavelength speed of coherent interfaces. While the industry is currently commercializing 1.6-Tb/s coherent optics, reaching the next interface speed doubling at 3.2 Tb/s remains a formidable challenge. As it becomes more challenging to scale the analog bandwidth of CMOS-based D/A converters (DACs) with the symbol rate, optical [1,2] or electronic multiplexing of lower-bandwidth signals is being actively investigated. While optical multiplexing has recently reached 400 GBaud [2], it requires multiple phase-locked optical carriers and parallel modulators; electronic high-symbol-rate generation needs only a single laser and a single electro-optical (E/O) modulator, minimizing transceiver cost and footprint. Fig. 1 summarizes electronically generated coherent transmissions with net bit rates (NBRs) beyond 1.6 Tb/s per wavelength [3–15]: with electronic symbol rates capped near 250 GBaud, results have plateaued in the 2-Tb/s class, e.g. 2.52 Tb/s at 248 GBaud [15]. In contrast, the bandwidth of E/O modulators has far exceeded what state-of-the-art electronics can offer, with various modulator technologies demonstrating E/O bandwidth beyond 200 GHz [16]. In 2025, we demonstrated all-electronic signal generation at 440 GBaud using a 4-band digital-band-interleaved (DBI) DAC [17]. That work was, however, confined to one-dimensional (1-D) amplitude-shift keying (ASK) from a single null-biased Mach–Zehnder modulator (MZM), delivering 1.04 Tb/s in one dimension. Transitioning

such a >200-GHz electrical (>400-GHz optical) signal from 1-D modulation to coherent I/Q signaling is not trivial. The main challenge comes from managing the I/Q mismatch at both transmitter and receiver. While a delay or an average power difference between I and Q can be easily corrected, the *frequency-dependent* part is not. It arises in nearly every component and may grow with frequency: the two arms of an I/Q modulator diverge in E/O response toward higher frequencies, and the two DBI-DAC chains differ band by band, both must be managed beyond 200 GHz. With 1-D modulation, a frequency-dependent imperfection of the chain is merely a linear distortion that an equalizer can remove; with two tributaries forming a complex signal, the difference between their imperfections becomes frequency-dependent image crosstalk that can severely limit the system SNR. Besides, at the receiver, the signal exceeds the bandwidth of available coherent receivers and is detected in two spectral slices [17]. For a real-valued signal the slices are conjugate mirrors of one another, so a stitching error at most forfeits a power combining gain; for a complex signal they carry independent data, so any residual mismatch between them distorts information that exists in no other slice. Here, we build a pair of 226-GHz DBI-DACs with end-to-end matched amplitude responses from DAC to modulator arm, and a receiver DSP that feeds both slices of both polarizations into a single widely-linear 8×2 multiple-input multiple-output (MIMO) equalizer for joint slice combining, I/Q imbalance image cancellation and polarization demultiplexing, and demonstrate an all-electronic coherent transmission at 450 GBaud with a record single-wavelength NBR of 3.83 Tb/s.

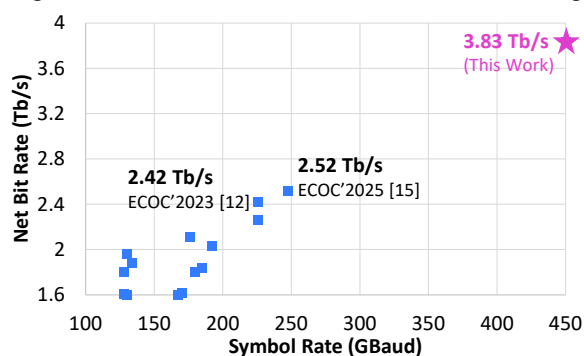


Fig. 1: Recently reported high-speed coherent transmissions ≥ 1.6 Tb/s/λ based on electronic high-symbol-rate generation.

The 450-GBaud I/Q Transmitter

We build a pair of DBI-DACs following the 4-band architecture of [17] with the analog bandwidth extended to 226 GHz; the band plan, intermediate-frequency (IF) ranges and radio-frequency (RF) local oscillators (LOs) are summarized in Fig. 3(i). Band 1 stays at baseband, while Bands 2–4 are up-converted by RF LOs at 88, 128 and 226.5 GHz, respectively. The four bands are combined by two power combiners and a diplexer (with a

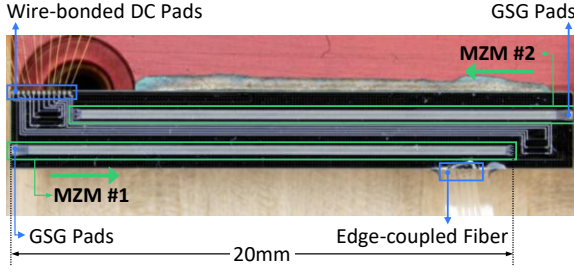


Fig. 2: Microscope image of the customized TFLN I/Q modulator. The I and Q MZMs are arranged in an anti-parallel configuration with GSG launch pads located on opposite edges.

crossover frequency of ~ 130 GHz) into a continuous band from DC to 226 GHz. The frequency responses and path delays of RF components are calibrated and pre-compensated by offline digital signal processing (DSP) as shown in Fig. 3(iii). Such pre-equalization also matches the I/Q chains to each other to minimize I/Q imbalance. Eight synchronized arbitrary waveform generator channels (Keysight M8199A/B) generate the baseband signals, and a bank of Micram DAC5 (followed by frequency multipliers) generates the desired LO frequencies, all driven by a common 64-GHz clock.

With only Ground-Signal-Ground (GSG) configuration available for our 450-GBaud signals, we customize a thin-film LiNbO₃ (TFLN) modulator with an anti-parallel electrode design (Fig. 2): the optical waveguides are routed such that the travelling-wave I and Q MZMs are oriented in opposite directions. This allows the two DBI-DAC outputs to be launched from opposite chip edges by two independent GSG probes, with sufficient mechanical clearance and no compromise in RF performance. With an electrode length of ~ 2 cm,

the single-polarization I/Q modulator has a 6-dB bandwidth above 110 GHz, a V_π of ~ 2 V (at 2 GHz), and an insertion loss of ~ 8 dB.

The light source is an external cavity laser (ECL) at 1550.12 nm with about 1-kHz linewidth, amplified to 22 dBm by an EDFA; with the I/Q signal pair turned on, the modulator output power is about -12 dBm. After amplification by a second EDFA, a polarization-division-multiplexing (PDM) signal is emulated by combining the signal with a decorrelated copy delayed by 10-m fiber (50 ns).

Coherent Detection and Receiver DSP

As the 450-GBaud signal exceeds the bandwidth of available opto-electronic receivers, we employ (as in [17]) optical arbitrary waveform measurement (OAWM) [18] with two coherent receivers each detecting half (~ 226 GHz) of the optical spectrum. We generate an E/O comb by driving an optical phase modulator (PM) with a 37.5-GHz RF tone and select two lines spaced at 225 GHz (6×37.5) by a wavelength selective switch (WSS) and route them to the two coherent receivers as LOs. The signal is sliced from the middle of its spectrum by a second WSS, and each half is routed to one receiver. Each receiver consists of a 90° optical hybrid, two balanced photodiodes (BPDs) and two real-time oscilloscope (RTO) channels sampling at 256 GSa/s with 113-GHz brick-wall bandwidth. Since a PDM signal would need two such OAWM setups and only one 4-channel RTO is available, the two polarizations are detected in a time-division multiplexing (TDM) manner [19]: the PDM signal is gated by an optical switch (SW) (4- μ s period, 50% duty cycle) and split by a polarization beam splitter (PBS), one

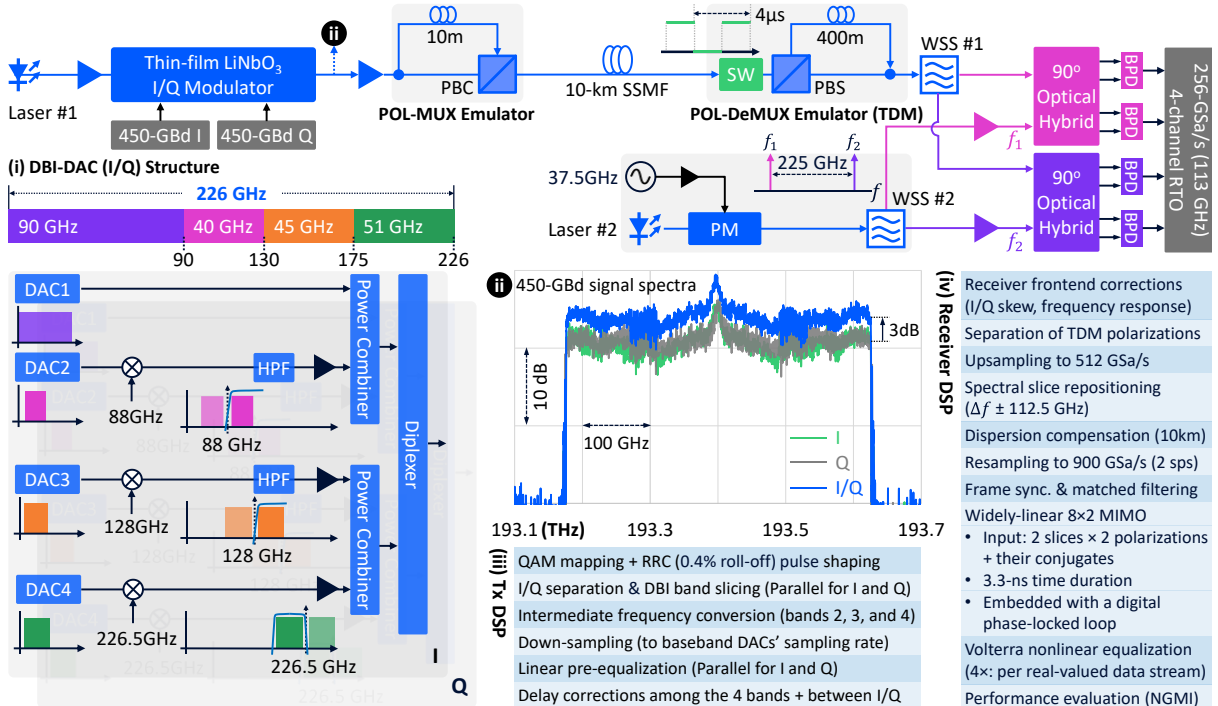


Fig. 3: Experimental setup for all-electronic 450-GBaud I/Q signal generation and OAWM coherent detection.

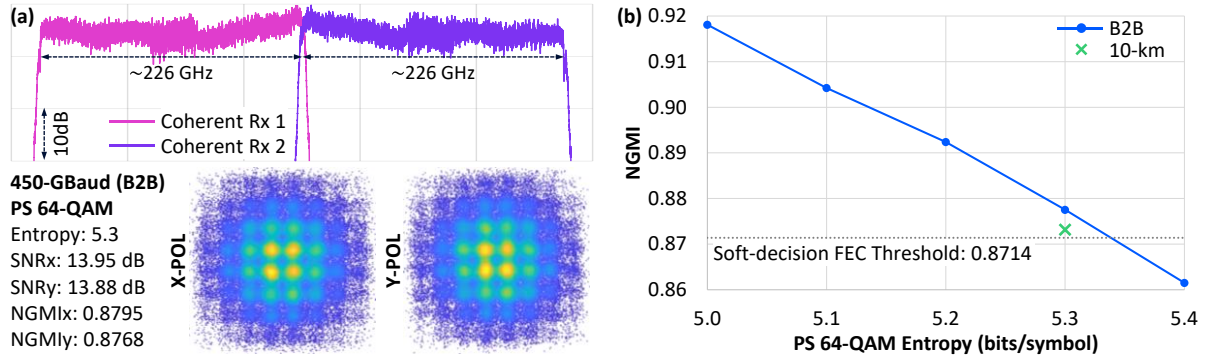


Fig. 4: (a) Reconstructed spectra of the two coherent receivers in OAWM and recovered constellations of the 450-GBaud PS 64-QAM signal at back-to-back (B2B) condition; (b) NGMI versus entropy of PS 64-QAM in B2B and after 10-km SSMF.

polarization is delayed by 400-m fiber (2 μ s), and the two paths are polarization-aligned and re-combined to be detected by the single-polarization OAWM receiver.

The receiver DSP mainly contains two stages as detailed in Fig. 3(iv). In stage 1, the entire 450-GHz spectrum is reconstructed by OAWM: the frequency-dependent I/Q imbalance of each coherent front-end is first compensated with pre-characterized responses; then the frequency offset Δf between the transmitter carrier and the comb seed is estimated, and the two slices are shifted by $\Delta f \pm 112.5$ GHz to their positions in the full-band spectrum. In stage 2, the 450-GBaud PDM signal is processed at 2 samples per symbol by MIMO equalization with an embedded phase-locked loop, followed by 4 parallel third-order Volterra post-filters (memory: 385, 25 and 13 symbols for the linear, quadratic and cubic kernels) applied separately to the I/Q outputs of each polarization to compensate nonlinearities arising in each DBI-DAC. Rather than adding the two slices into one waveform before equalization as for the ASK signal in [17], we feed both spectral slices into a single widely-linear 8×2 MIMO equalizer, whose eight inputs are the two polarizations of the two slices and their conjugates. Keeping the slices separate matters because the I/Q signal is not Hermitian symmetric: a mismatch between the two slices is no longer a bounded combining loss (for 1-D signals) but a direct distortion of the reconstructed signal. With the slices as MIMO inputs, this mismatch is absorbed adaptively by the equalizer. Its widely-linear structure additionally cancels the mirror-frequency image produced by the transmitter I/Q imbalance. One set of taps thus performs slice combining, image correction and polarization demultiplexing.

Results

The transmitted signal is 450-GBaud probabilistically shaped (PS) 64-QAM with root-raised cosine (RRC, 0.4% roll-off) pulse shaping, leading to an optical bandwidth of 451.8 GHz as shown in Fig. 3(ii). Fig. 3(ii) also compares the optical

spectra measured with only the I or only the Q signal applied, each representing the end-to-end response of one tributary from DBI-DAC to modulator arm. Their overlap across the 226-GHz band confirms that the two tributaries are closely matched after pre-equalization, a prerequisite for high-quality I/Q signaling. The combined I/Q spectrum is about 3 dB higher than either, as expected. Fig. 4(a) shows the corresponding pair of digital spectra of the two slices after the receiver front-end correction and OAWM spectral alignment, which together resemble the optical spectrum in Fig. 3(ii).

The PDM signal is evaluated in back-to-back (B2B) and after transmission over 10-km standard single-mode fiber (SSMF). We choose a forward error correction (FEC) scheme with a code rate (c) of 0.8262 that concatenates a soft-decision spatially-coupled low-density parity-check (LDPC) inner code (rate 5/6) and a hard-decision BCH outer code (rate 0.9915), with a normalized generalized mutual information (NGMI) threshold of 0.8714 [20]. The NBR is calculated as

$$NBR = 2B(H - (1 - c) \log_2 M)$$

where the factor of 2 comes from PDM, B is the symbol rate, H is the entropy (bits/symbol), and M is the QAM order. In Fig. 4(b), we vary the entropy of PS 64-QAM signals and measure their NGMI. The highest entropy that yields an NGMI higher than the threshold is 5.3 bits/symbol after 10-km SSMF, leading to an NBR of 3.83 Tb/s. This corresponds to 0.96 Tb/s per modulation dimension, close to the 1.04 Tb/s of the 1-D ASK signal in [17]. A reduction is expected, since I/Q tributaries share optical power, lowering the per-dimension optical SNR (OSNR), and residual I/Q imbalance and slice mismatch have no counterpart for a real-valued ASK signal.

Conclusions

With a net bit rate of 3.83 Tb/s, we have shown that all-electronic signal generation can bring a single wavelength into the 3.2-Tb/s class without relying on optical multiplexing. The results also imply that I/Q modulators developed for the 1.6-Tb/s generation may evolve smoothly to support future higher-speed coherent interfaces.

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